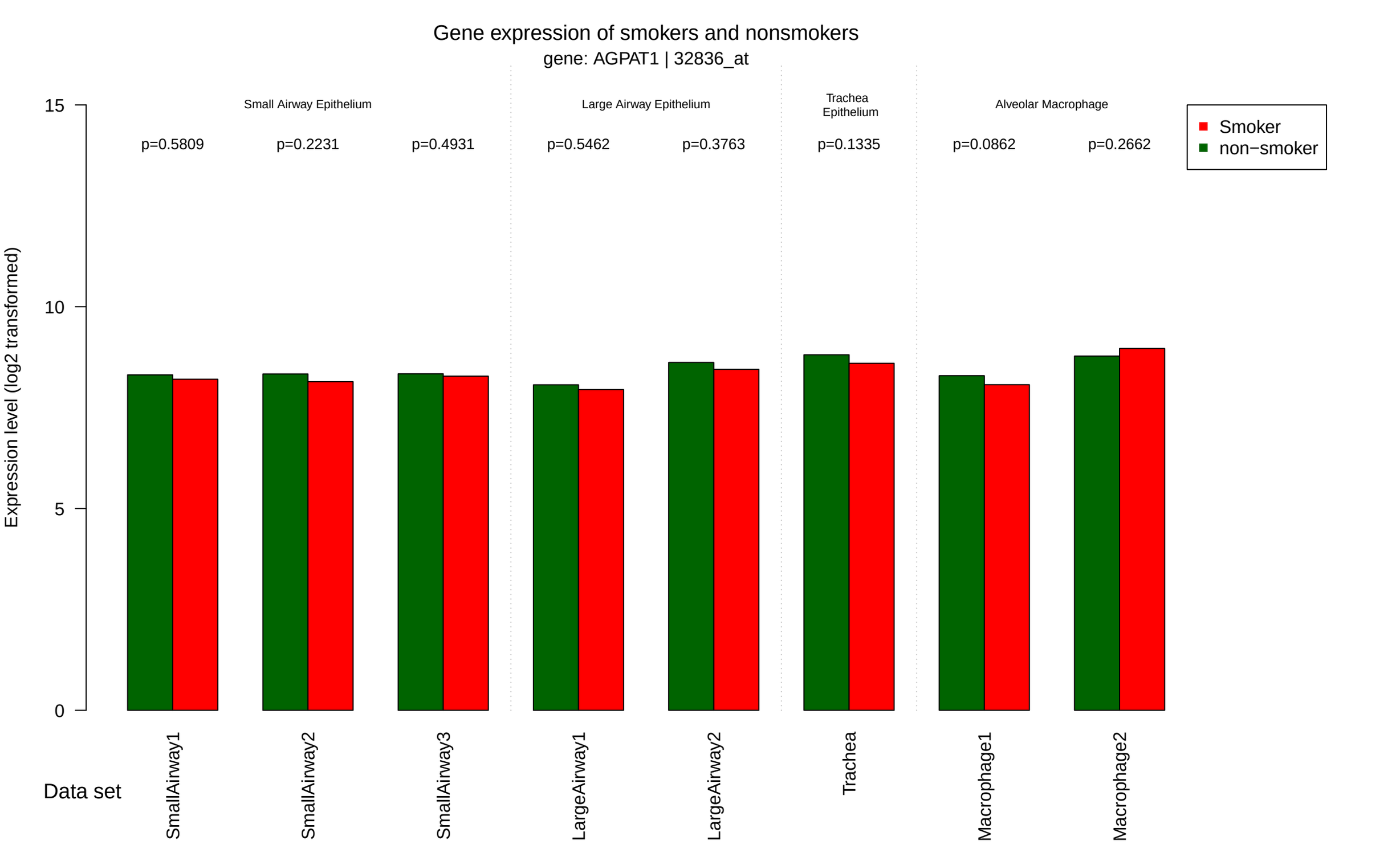
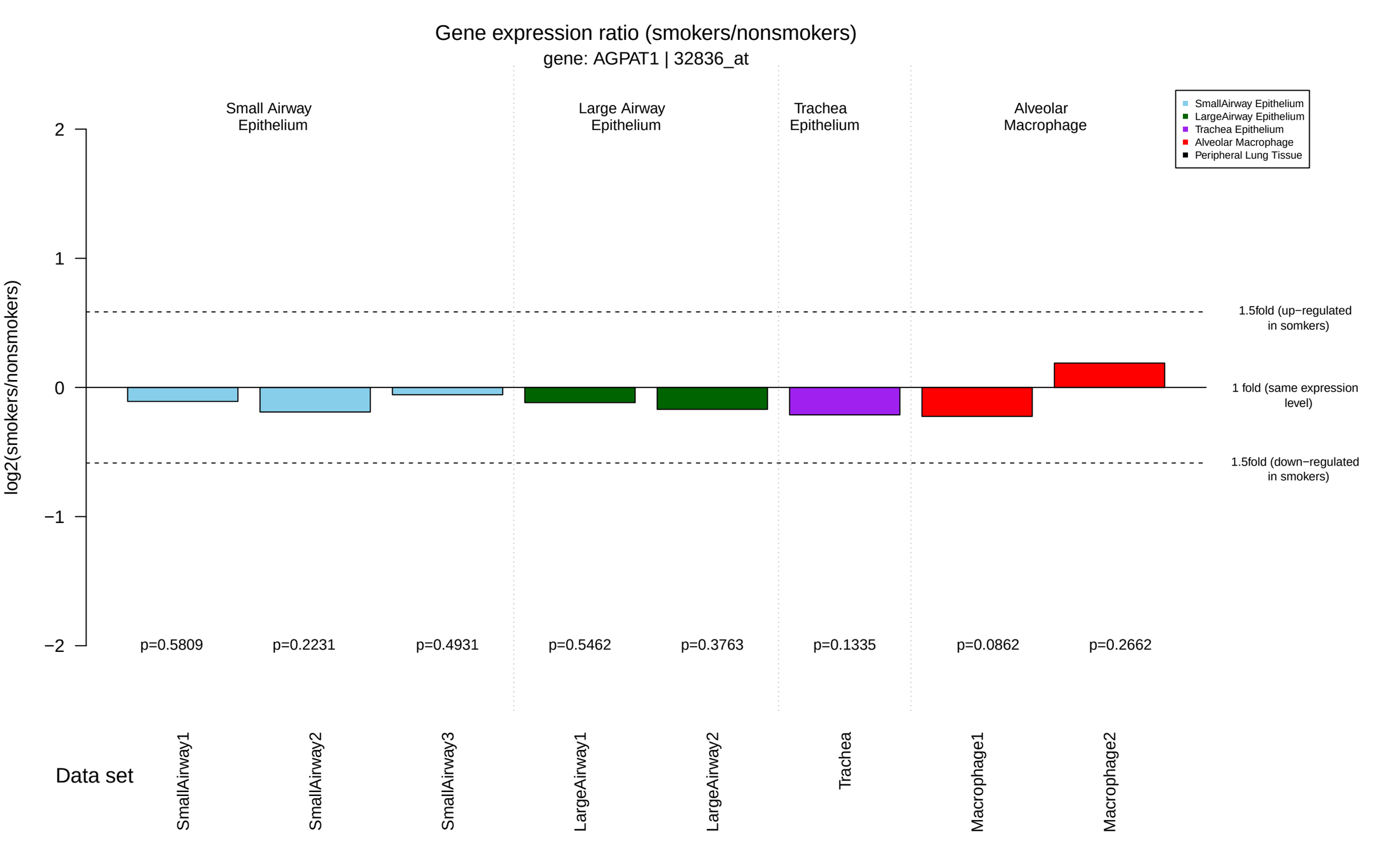
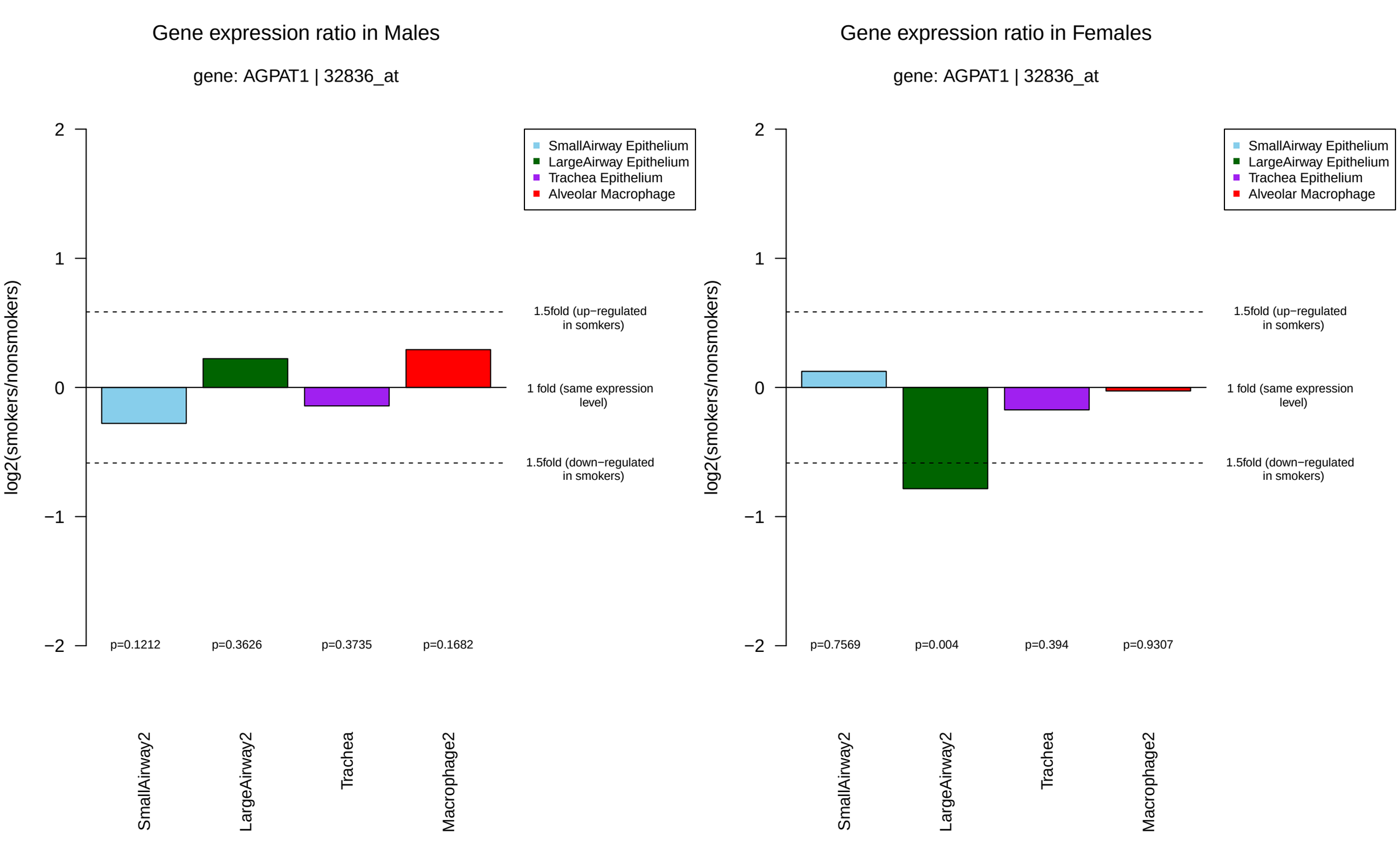
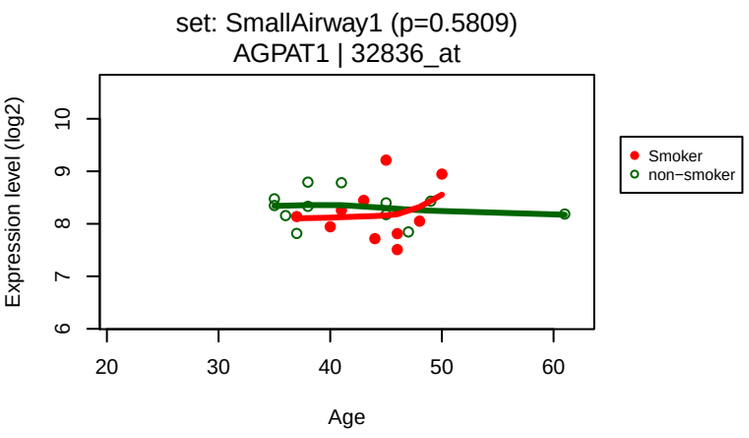
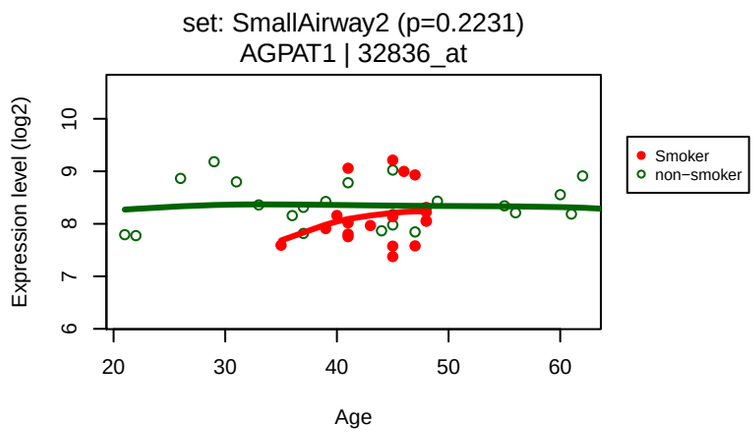
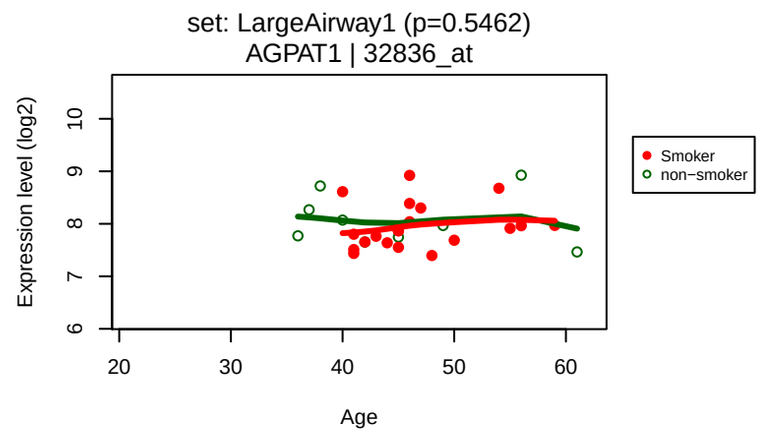

Gene expression of smokers and nonsmokers
gene: AGPAT1 | 32836_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5809
p=0.2231
p=0.4931
p=0.5462
p=0.3763
p=0.1335
p=0.0862
p=0.2662
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

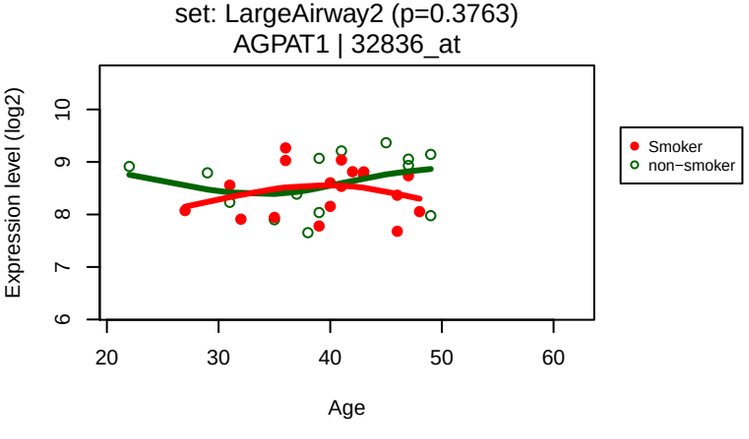

Gene expression ratio (smokers/nonsmokers)
gene: AGPAT1 | 32836_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5809
p=0.2231
p=0.4931
p=0.5462
p=0.3763
p=0.1335
p=0.0862
p=0.2662
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

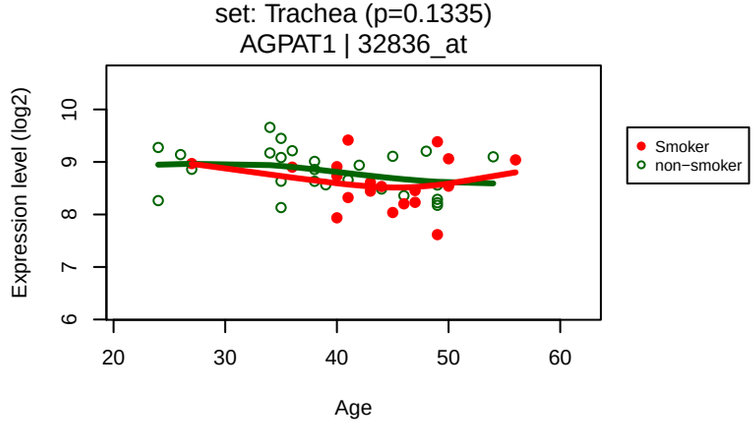

Gene expression ratio in Males
gene: AGPAT1 | 32836_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1212
p=0.3626
p=0.3735
p=0.1682
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AGPAT1 | 32836_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7569
p=0.004
p=0.394
p=0.9307
SmallAirway2
LargeAirway2
Trachea
Macrophage2

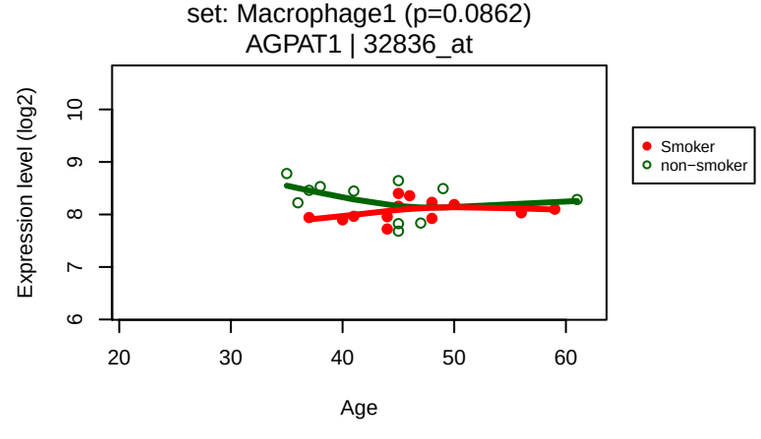

set: SmallAirway1 (p=0.5809)
AGPAT1 | 32836_at
Expression level (log2)
Age
Smoker
non−smoker

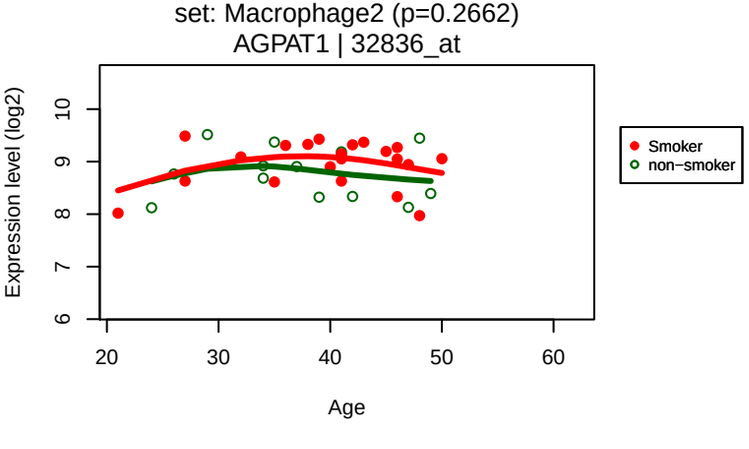

set: SmallAirway2 (p=0.2231)
AGPAT1 | 32836_at
Expression level (log2)
Age
Smoker
non−smoker

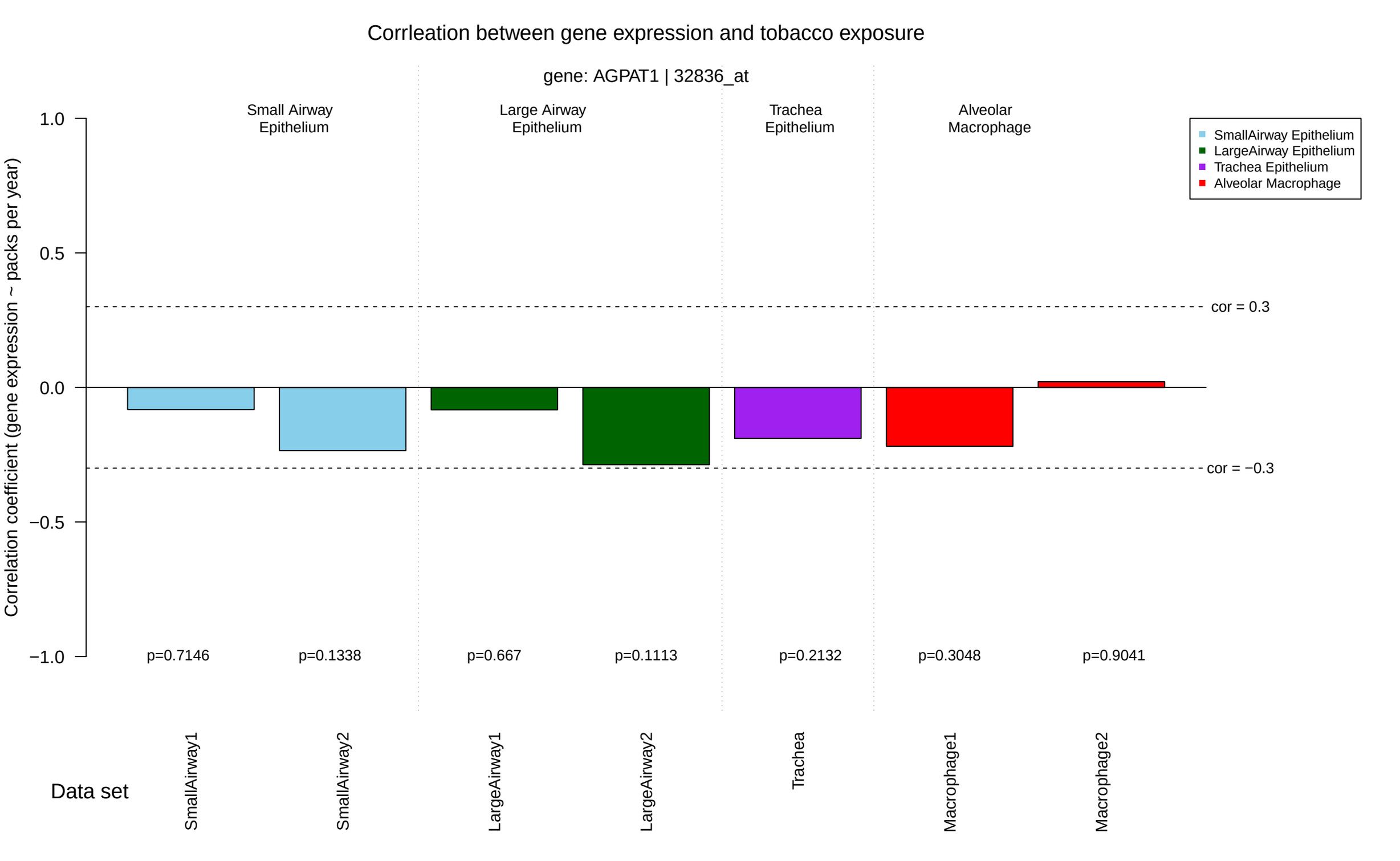

set: LargeAirway1 (p=0.5462)
AGPAT1 | 32836_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3763)
AGPAT1 | 32836_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1335)
AGPAT1 | 32836_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0862)
AGPAT1 | 32836_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2662)
AGPAT1 | 32836_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AGPAT1 | 32836_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7146
p=0.1338
p=0.667
p=0.1113
p=0.2132
p=0.3048
p=0.9041
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2